
HW3

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4.1 Answer each of the following:

- Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum e_i^2 = 182.85$, find R^2 .

$$a. SST = \sum (y_i - \bar{y})^2 \Rightarrow SST = \sum y_i^2 - N\bar{y}^2$$

$$SST = 7825.94 - 20 \times (19.21)^2 = 445.458$$

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8428$$

$$b. \hat{\sigma}^2 = \frac{SSE}{N-2}, R^2 = \frac{SSR}{SST} = 0.7911 \Rightarrow SSR = 574.291134$$

$$SSE = SST - SSR = 151.648866 - 0$$

$$\hat{\sigma}^2 = \frac{0}{18} = 8.424937$$

$$c. R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105 \#$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 e_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- the predicted value of y for $x_0 = 4$.
- the $se(f)$ corresponding to part (a).
- a 95% prediction interval for y given $x_0 = 4$.
- a 99% prediction interval for y given $x_0 = 4$.
- a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

$$a. \hat{y}_0 = 1.2 + 0.8 \times 4 = 4.4$$

$$b. \widehat{Var}(f) = 1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right] = 2.52$$

$$se(f) = \sqrt{2.52} = 1.5875$$

$$c. tc = t_{(0.975, 3)} = 3.1824, \hat{y}_0 \pm tc \cdot se(f) = 4.4 \pm 3.1824 \times 1.5875$$

$$= [-0.6520, 9.4520]$$

$$d. tc = t_{(0.995, 3)} = 5.8409, \hat{y}_0 \pm tc \cdot se(f) = 4.4 \pm 5.8409 \times 1.5875$$

$$= [-4.8722, 13.6722]$$

$$e. \bar{x} = 1, \hat{y}_0 = b_1 + b_2 \bar{x} = 1.2 + 0.8(1) = 2$$

$$\widehat{\text{Var}}(\hat{y}) = \hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right) = 1.2 \left(1 + \frac{1}{5} + \frac{(-1)^2}{10} \right) = 1.44$$

$$\text{se}(\hat{y}) = \sqrt{1.44} = 1.2$$

$$t_c = t_{\text{conf}(5,3)} = 3.1824, \hat{y}_0 \pm t_c \text{se}(\hat{y}) = 2 \pm 3.1824 \times 1.2$$

$$= 2 \pm 3.81888$$

$$= [-1.81888, 5.81888]$$

$\Rightarrow (e)$ is shorter than (c) #